

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

FRIDAY, 20 MAY 2022

TIME : 2.30 PM – 4.30 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** questions in **Section B**, marks will only be awarded for the **FIRST** questions in the order the candidate submits the answers.

No calculator is allowed for this paper.

Answer questions only in the answer booklet provided.

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**SECTION A**

Q1. (a) The Basic Structure of a JavaFX Program questions.

- (i) Explain the three major components used in JavaFX application. (3 marks)
- (ii) Given three radio buttons; **rbRed**, **rbYellow** and **rbGreen**. Write code to make radio buttons mutually exclusive grouping. (4 marks)
- (iii) Write code fragments to add two nodes, named *node1* and *node2*, into a pane named *pane*. (1 mark)

(b) Fill in the code below in the missing part to display the **mouse point location** when the mouse is **pressed** in the pane. (2 marks)

```
public class Test extends Application {
    @Override
    public void start(Stage primaryStage) {
        Pane pane = new Pane();

        // missing part

        Scene scene = new Scene(pane, 200, 250);
        primaryStage.setTitle("Test");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

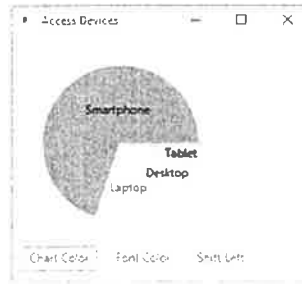
(c) Write a simple JavaFX application that uses a pie chart to display the percentages of the access devices such as *Smartphone*, *Laptop*, *Desktop*, and *Tablet*, and contains **three (3)** buttons for manipulating colors and locations of the pie chart, as shown in **Figure 1.1**.

1st button: Change the color of the pie chart (color selection is based on your own preference).

2nd button: Change the color for the fonts in the pie chart (color selection is based on user preference).

3rd button: Move the pie chart and fonts to the right side by **10 pixels** (for each click).

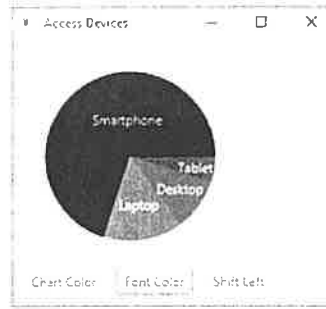
Complete the program by filling in the parts: (i), (ii), (iii), (iv) and (v).

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (c) (Continued)**

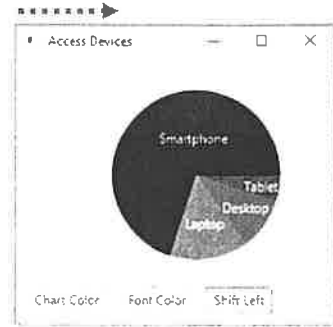
(a) The initial Screen.



(b) Button "Chart Color" is pressed.



(c) Button "Font Color" is pressed.



(d) Button "Shift Left" is pressed.

Figure 1.1

```
public class Q1_c extends Application {

    public void start(Stage primaryStage) {

        // Pane pane for placing the arcs and texts
        Pane pane = new Pane();
        Arc[] arcs = new Arc[4];
        Text[] txts = new Text[4];

        // first arc
        arcs[0] = new Arc(100, 100, 80, 80, 0, 360 * 0.7);
        arcs[0].setFill(Color.LIGHTCORAL);
        arcs[0].setType(ArcType.ROUND);
        txts[0] = new Text(65, 70, "Smartphone");

        // second arc
        arcs[1] = new Arc(100, 100, 80, 80, 360 * 0.7, 360 * 0.15);
        arcs[1].setFill(Color.LIGHTGOLDENRODYELLOW);
        arcs[1].setType(ArcType.ROUND);
        txts[1] = new Text(90, 150, "Laptop");

        // third arc
        arcs[2] = new Arc(100, 100, 80, 80, 360 * 0.85, 360 * 0.10);
        arcs[2].setFill(Color.LIGHTGREEN);
        arcs[2].setType(ArcType.ROUND);
        txts[2] = new Text(125, 135, "Desktop");

        // refer to (i)
```

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (c) (Continued)**

```

// refer to (ii)

// HBox pane for holding the three buttons
HBox hBox = new HBox();
hBox.setSpacing(10);
hBox.setAlignment(Pos.BASELINE_LEFT);
Button btChartCol = new Button("Chart Color");
Button btFontCol = new Button("Font Color");
Button btPosition = new Button("Shift Left");
hBox.getChildren().addAll(btChartCol, btFontCol,
btPosition);

// refer to (iii)

// Create a border pane and place pane and hBox in the
border pane
BorderPane borderPane = new BorderPane();
borderPane.setCenter(pane);
borderPane.setPadding(new Insets(10));
borderPane.setBottom(hBox);
BorderPane.setAlignment(hBox, Pos.BOTTOM_CENTER);

// refer to (iv)

}

// refer to (v)

}

```

- (i) Create the last arc for *Tablet* that takes 5% and displays it using LIGHTSALMON color. Set the arc type to ArcType.ROUND. (2 marks)
- (ii) Add four arcs and texts into the pane. (2 marks)
- (iii) Add the event handling code to 3 buttons; btChartCol, btFontCol and btPosition using Lambda expression. Perform the respective actions as mentioned above (see Figure 1.1 (b) – (d)) when the buttons are pressed. (8 marks)
- (iv) Create a scene of size (300,250) to host the border pane, place the scene into the stage, and show the stage. (2 marks)
- (v) Add main() to launch the application. (1 mark)

[Total : 25 marks]

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

Q2. (a) The following questions are based on File Input and Output,

(i) What are the **main differences** between text I/O and binary I/O? (4 marks)

(ii) Write a program to create a file named "*temp.dat*" if it does not exist. Append new data to it if it already exists. Uses binary I/O to write **ten byte values** from 1 to 10 to a file named "*temp.dat*" and reads them back from the file.

Output: 1 2 3 5 6 7 8 9 10 (11 marks)

(b) The following question is based on Java Database Programming,

(i) Given the URL "jdbc:ucanaccess://Drive:/DatabaseDir/DatabaseLab" for an Access database that contains a table, named **Enrollment**, write a complete Java Database program (JDBC) to connect the database, query results and display all the column names and contents resulting from the **Enrollment table**. Hint: **ResultSetMetaData** (10 marks)

Output:

Driver loaded

Successfully connected to database

ssn	courseld	dateRegistered	grade
1222	11120	2022-01-03	A
1222	11121	2022-01-03	B
1222	11121	2022-01-03	B

[Total : 25 marks]

Q3. (a) Explain the difference between checked exception and unchecked exception in terms of exception handling. List an exception error for each type of exception. (4 marks)

(b) What does the JVM do when a runtime error occurs? How do you catch a runtime error? (4 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICESQ3. (Continued)

- (c) Suppose methodY(double d) causes an exception in the following try-catch block:

```
public void methodQ4()throws Exception {
    try {
        methodY(0.5);
    }
    catch (Exception ex) {
        System.out.println(ex);
        throw ex;
    }
    finally {
        System.out.println("Finally block executed");
    }
    System.out.println("Program continue");
}
```

Provide the code for methodY(double d), in which it should throw an Exception object with error message "Input argument must >= 1" when the input argument d is smaller than 1.0. (4 marks)

- (d) What is the output of the following code? (2 marks)

```
public class Test {
    public static void main(String[] args) {
        try {
            int value = 30;
            if (value < 40)
                throw new Exception("value is too small");
        }
        catch (Exception ex) {
            System.out.println(ex.getMessage());
        }
        System.out.println("Continue after the catch block");
    }
}
```

- (e) What would be the output if the statement (int value = 30;) in Q3(d) is replaced with this statement: int value = 50; (1 mark)
- (f) The following questions are based on thread and Multithreading,
- (i) What is the difference between **Process** and **Thread**? (2 marks)
- (ii) How do you **define** a Task class? How do you **create** a thread for a Task object? (2 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q3. (Continued)**

- (g) Analyze the following code:

```
public class Test extends Thread {  
    public static void main(String[] args) {  
        Test t = new Test();  
        t.start();  
        t.start();  
    }  
  
    public void run() {  
        System.out.println("test");  
    }  
}
```

Is there any error? Briefly provide your reason.

(3 marks)

- (h) How do you create a thread pool with three fixed threads? How do you submit a task to a thread pool?

(3 marks)

[Total : 25 marks]

SECTION B

This section consists of **TWO (2)** questions. Answer **ONLY ONE (1)** question.

- Q4. (a) Define the following terms with regards to object-oriented programming.

(i) Class (2 marks)

(ii) Object (2 marks)

(iii) Constructor (2 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICESQ4. (Continued)

- (b) Answer the following questions based on the Java code below:

```
1. public class ClassQ4
2. {
3.     public static void main(String[] args)
4.     {
5.         Course course = new Course();
6.         System.out.println(course.numStudents);
7.         int number = 10;
8.         setNumStudent(number, course);
9.         System.out.println(number);
10.        System.out.println(course.numStudents);
11.    }
12.
13.    public static void setNumStudent(int number,
Course c)
14.    {
15.        c.numStudents = number;
16.        number = 0;
17.    }
18. }
19.
20. class Course
21. {
22.     public int numStudents;
23. }
```

- (i) Explain the differences between pass by value and pass by reference in terms of argument value and changes made on the parameter inside a method? (4 marks)
- (ii) What are the outputs of lines 6, 9, and 10? (3 marks)
- (c) What are the **THREE (3)** requirements for a class to be immutable? (3 marks)
- (d) Analyze the following code:

```
public class A {
    private int[] values;

    public int[] getValues() {
        return values;
    }
}
```

Is class A an **immutable** class? Justify your answer.

(2 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICESQ4. (Continued)

- (e) Regarding to the programming errors,
- (i) Explain the meaning of syntax errors, runtime errors, and logic errors. (3 marks)
 - (ii) What will happens if the program needs to read integers, but the user entered strings? (2 marks)
 - (iii) Suppose you want to write a program to compute the perimeter of a circle, but you accidentally write the program to compute the area of a circle. What kind of error is and how do you avoid it? (2 marks)
- [Total : 25 marks]

- Q5. (a) What is the difference between **abstract** class and **interface**? (4 marks)
- (b) Create an *abstract* class named **BookStore** to represent an item in a library, which contains the following requirements. Write your answer in a single Java class definition.
- (i) Two private `String` data fields, named *title* and *barCode*. (2 marks)
 - (ii) A constructor with signature `BookStore(String barCode)` to initialize the *barCode* data field and set *title* to "no title". Default constructor is not required. (2 marks)
 - (iii) Two accessor methods for the two data fields, and one mutator method for the *title* data field. (3 marks)
- (c) **Book** and **Comic** are two concrete subclasses of the **BookStore** class.
- (i) The **Book** class has a private `String` data field named *author* and a constructor for creating book instance with the provided barcode and author name. Write the class definitions for **Book**. (3 marks)
 - (ii) The **Comic** class has a private `int` data field named *issue* and a constructor for creating magazine instance with the provided barcode and issue number. Write the class definitions for **Comic**. (3 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICESQ5. (Continued)

- (d) A rental store has different books and comics. Create the **RentalStore** class which contains the following requirements. Write your answer in a single Java class definition.
- (i) A private `ArrayList` data field of `BookStore` named *itemList* for storing all the books and comics that the rental store has. (2 marks)
 - (ii) A method named `addItem(BookStore item)` that adds a new rental store item to *itemList*. (1 mark)
 - (iii) A method named `removeItem(BookStore item)` that removes a rental store item from *itemList*. (1 mark)
 - (iv) A method named `getNumberOfItems()` that returns the total number of store items that the rental store currently has. (1 mark)
 - (v) A method named `getNumberOfBooks()` that returns the total number of books in the rental store. (3 marks)
- [Total : 25 marks]
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